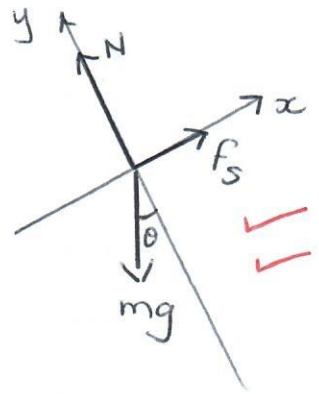
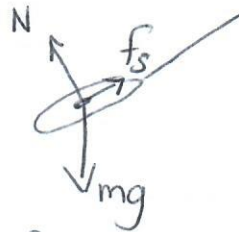


1. The coefficient of static friction between Teflon and scrambled eggs is about 0.04. Determine the smallest angle from the horizontal that will cause the eggs to slide across the bottom of a Teflon-coated pan? Start from the application of Newton's 2nd law, and show all your steps. *Die statiese wrywingskoëffisient tussen Teflon en roereier is omtrent 0.04. Bepaal die kleinste hoek tussen die pan en die horisontaal wat sal veroorsaak dat die eiers oor die bodem van die Teflon-bedekte pan sal gly. Begin met die toepassing van Newton se 2de wet, en wys al jou stappe.* (5)
2. What is the smallest radius of an unbanked (flat) track around which a cyclist can travel at a speed of 29 km/h if the μ_s between tires and track is 0.32? *Wat is die kleinste radius van 'n plat baan waarop 'n fietsryer kan ry teen 'n speed van 29 km/h indien die μ_s tussen die bande en die baan 0.32 is?* (5)

Tutorial 5-1

1. $\mu_s = 0.04$



$$\Sigma F_y = ma_y = 0 = N - mg \cos \theta$$
$$N = mg \cos \theta. \checkmark$$

$$\Sigma F_x = ma_x = 0 = f_s - mg \sin \theta = \mu_s N - mg \sin \theta$$
$$= \mu_s mg \cos \theta - mg \sin \theta \checkmark$$

$$\tan \theta = \mu_s = 0.04$$

$$\theta = 2.3^\circ \checkmark$$

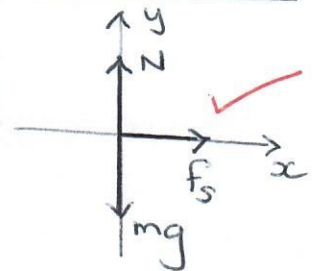
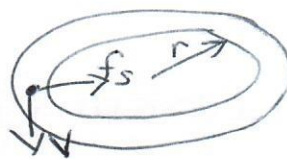
An angle just greater than 2.3° will cause the eggs to slide

2. $\mu_s = 0.32$

$$v = 29 \text{ km/h}$$

$$= \frac{29 \text{ km}}{\text{h}} \times \frac{1 \text{ h}}{60 \text{ min}} \times \frac{1 \text{ min}}{60 \text{ s}} \times \frac{1000 \text{ m}}{1 \text{ km}}$$

$$= 8.05 \text{ m/s} \checkmark$$



$$\Sigma F_y = ma_y = 0 = N - mg$$
$$\therefore N = mg \checkmark$$

$$\Sigma F_x = ma_c = f_s = \mu_s N$$

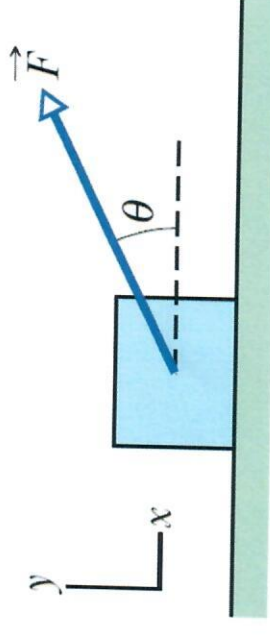
$$\therefore a_c = \mu_s \frac{mg}{m} = \frac{v^2}{r} \checkmark$$

$$\frac{v^2}{r} = \mu_s g$$

$$r = \frac{v^2}{\mu_s g} = \frac{(8.05 \text{ m/s})^2}{0.32 (9.8 \text{ m/s}^2)}$$

$$r = 20.66 \text{ m} \checkmark$$

1. The figure shows an initially stationary block of mass m on a floor. A force of magnitude $0.500mg$ is then applied at upward angle $\theta = 20^\circ$. What is the magnitude of the acceleration of the block across the floor if the friction coefficients are $\mu_s = 0.600$ and $\mu_k = 0.500$.



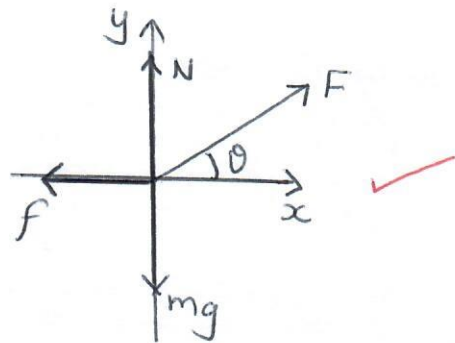
(5)

2. During an Olympic bobsled run, the Jamaican team makes a turn of radius 7.6 m at a speed of 96.6 km/h . What is their acceleration in terms of g ?
Tydens 'n Olimpiese "bobsled" wedloop, het Jamaika se span in 'n draai met 'n radius van 7.6 m gegaan teen 'n spoed van 96.9 km/h en suksesvol anderkant uitgekom. Wat is die versnelling in terme g ?

(5)

Tutorial Test 5-2

$$1. \sum F_y = ma_y = 0 = N + F \sin \theta - mg$$
$$N = mg - F \sin \theta \quad \checkmark$$



$$\sum F_x = ma_x = F \cos \theta - f_k$$
$$a_x = \frac{F \cos \theta - \mu_k N}{m} \quad \checkmark \quad \text{--- ①}$$

$$f_{s, \max} = \mu_s N = \mu_s (mg - F \sin \theta) \quad \checkmark$$
$$= 0.600 (mg - 0.500 mg \sin 20^\circ)$$
$$= 0.600 mg (1 - 0.500 \sin 20^\circ)$$
$$= 0.497 mg \quad \checkmark$$

$$F \cos \theta = 0.500 mg \cos 20^\circ = 0.470 mg \quad \checkmark$$

From ① we see $F \cos \theta < f_{s, \max}$.

Block does not move, $a = 0 \quad \checkmark$

$$2. \quad v = 96.6 \frac{\text{km}}{\text{h}} \times \frac{1000 \text{ m}}{1 \text{ km}} \times \frac{1 \text{ hr}}{3600 \text{ s}} = 26.8 \text{ m/s} \quad \checkmark$$

$$a = \frac{v^2}{r} = \frac{(26.8 \text{ m/s})^2}{7.6 \text{ m}} = 94.5 \text{ m/s}^2 \quad \checkmark$$

$$a = \left(\frac{a}{g} \right) g = \left(\frac{94.5 \text{ m/s}^2}{9.8 \text{ m/s}^2} \right) g \quad \checkmark = 9.6g \quad \checkmark$$

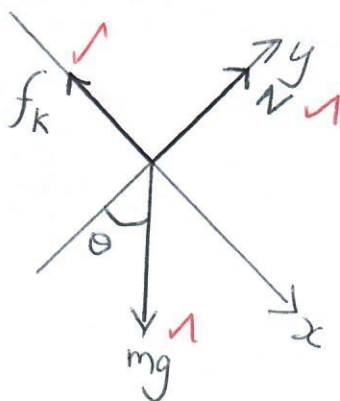
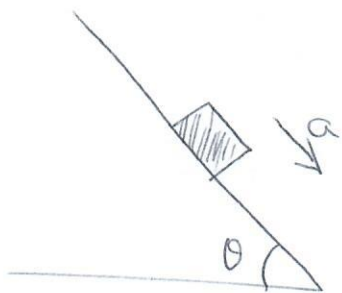
1. A box of canned goods slides down a ramp from street level into the basement of a grocery store with acceleration 0.75 m/s^2 directed down the ramp. The ramp makes an angle of 40° with the horizontal. What is the coefficient of kinetic friction between the box and the ramp?

'n Boks ingelegde goedere gly by 'n skuins vlak af van straatvlak tot in die kelder van 'n winkel teen 'n versnelling van 0.75 m/s^2 gerig langs die skuins vlak af. Die skuinsvlak maak 'n hoek van 40° met die horisontaal. Wat is die kinetiese wrywingskoëffisiënt tussen die boks en die skuinsvlak. (6)
2. A cyclist travels in a circle of radius 25.0 m at a constant speed of 9.00 m/s . The cyclist's mass is 85.0 kg . Calculate the magnitude of the force of friction on the bicycle from the road.

'n Fietsryer ry in 'n sirkel met 'n radius van 25.0 m teen 'n konstante spoed van 9.00 m/s . Die fietsryer het 'n massa van 85.0 kg . Bereken die grootte van die wrywingskrag op die fiets wat deur die pad uitgeoefen word. (4)

Tutorial Test 5-3

1.



✓ Axes.

$$\Sigma F_y = ma_y = 0 = N - mg \cos \theta \quad \checkmark$$
$$N = mg \cos \theta$$

$$\Sigma F_x = ma_x = mg \sin \theta - f_k = mg \sin \theta - \mu_k N \quad \checkmark$$

$$\mu_k = \frac{-ma + mg \sin \theta}{mg \cos \theta} \quad \checkmark$$

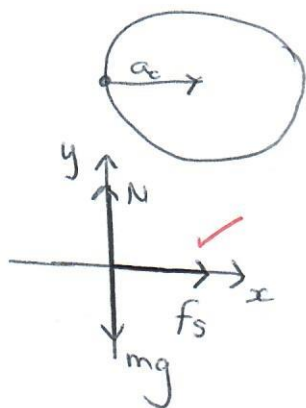
$$\mu_k = \frac{-a + g \sin \theta}{g \cos \theta} = \frac{-0.75 \text{ m/s}^2 + 9.8 \text{ m/s}^2 \sin 40^\circ}{9.8 \text{ m/s}^2 \cos 40^\circ}$$

$$\mu_k = 0.74 \quad \checkmark$$

2. $r = 25.0 \text{ m}$

$$v = 9.00 \text{ m/s}$$

$$m = 85.0 \text{ kg}$$



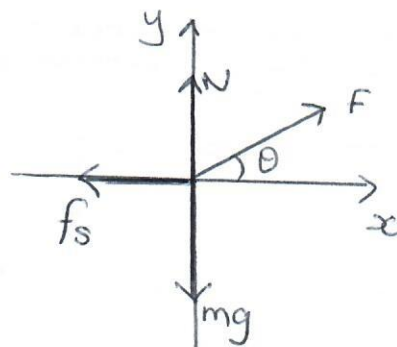
$$\Sigma F_x = ma_c = f_s = m \frac{v^2}{r} \quad \checkmark$$

$$f_s = 85.0 \text{ kg} \left(\frac{(9.00 \text{ m/s})^2}{25.0 \text{ m}} \right) \quad \checkmark$$

$$= 275 \text{ N} \quad \checkmark$$

1. A 68 kg crate is dragged across a floor by pulling on a rope attached to the crate and inclined 15° above the horizontal. (a) If the coefficient of static friction is 0.50, what minimum force magnitude is required from the rope to start the crate moving?
'n 68 kg krat word oor die vloer getrek deur 'n tou daaraan vas te maak en dit teen 15° bo die horisontaal te trek. As die statiese wrywingskoëffisiënt 0.5 is, wat is die grootte van die minimum krag op die tou wat nodig is om die krat te laat beweeg? (5)
2. A cat dozes on a stationary merry-go-round, at a radius of 5.4 m from the center of the ride. Then the operator turns on the ride and brings it up to its proper turning rate of one complete rotation every 6.0 s. What is the least coefficient of static friction between the cat and the merry-go-round that will allow the cat to stay in place, without sliding?
'n Kat slaap op 'n rondomtalie, op 'n afstand van 5.4 m van die middelpunt af. Die operateur skakel die rondomtalie aan en die normale draai tempo is een omwenteling elke 6.0 s. Was is die kleinste by die statiese wrywingskoëffisiënt tussen die kat en die rondomtalie wat sal toelaat dat die kat bly lê sonder om af te gly. (5)

Tutorial Test 5-4



$$\Sigma F_y = ma_y = 0 = N + F \sin \theta - mg$$

$$N = mg - F \sin \theta$$

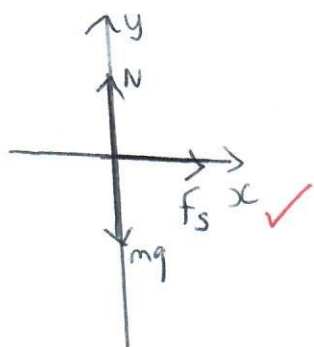
$$\Sigma F_x = ma_x = 0 = F \cos \theta - \mu_s N$$
$$= F \cos \theta - \mu_s mg + \mu_s F \sin \theta$$

$$F = \frac{\mu_s mg}{(\cos \theta + \mu_s \sin \theta)}$$
$$= \frac{0.5 (68 \text{ kg}) (9.8 \text{ m/s}^2)}{\cos 15^\circ + 0.5 (\sin 15^\circ)}$$
$$= 382 \text{ N}$$

2 $r = 5.4 \text{ m}$

$$T = 6.0 \text{ s}$$

$$v = \frac{\text{distance}}{\text{time}} = \frac{2\pi r}{T} = 5.65 \text{ m/s}$$



$$\Sigma F_y = ma_y = 0 = N - mg$$
$$\therefore N = mg$$

$$\Sigma F_x = ma_c = m \frac{v^2}{r} = fs = \mu_s N$$

$$\mu_s = \frac{mv^2}{mgr} = \frac{v^2}{gr}$$

$$\mu_s = 0.60$$

1. The coefficient of static friction between Teflon and scrambled eggs is about 0.06. Determine the smallest angle from the horizontal that will cause the eggs to slide across the bottom of a Teflon-coated pan?

Die statiese wrywingskoëfisiënt tussen Teflon en roereier is omtrent 0.06. Bepaal die kleinste hoek tussen die pan en die horisontaal wat sal veroorsaak dat die eiers oor die bodem van die Teflon-bedekte pan sal gly.

(7)

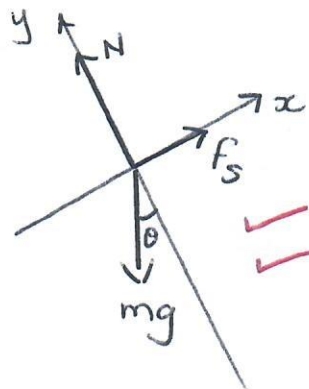
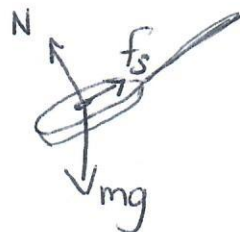
2. During an Olympic bobsled run, the Jamaican team makes a turn of radius 7.6 m at a speed of 96.6 km/h. What is their acceleration in terms of g?

Tydens 'n Olimpiese "bobsled" wedloop, het Jamaika se span in 'n draai met 'n radius van 7.6 m gegaan teen 'n spoed van 96.9 km/h en suksesvol anderkant uitgekom. Wat is die versnelling in terme g?

(3)

Tutorial 5-5

1. $\mu_s = 0.04$



$$\Sigma F_y = ma_y = 0 = N - mg \cos \theta$$
$$N = mg \cos \theta. \checkmark$$

$$\Sigma F_x = ma_x = 0 = f_s - mg \sin \theta = \mu_s N - mg \sin \theta$$
$$= \mu_s mg \cos \theta - mg \sin \theta \checkmark$$

$$\tan \theta = \mu_s = 0.04$$

$$\theta = 3.4^\circ \checkmark$$

An angle just greater than 3.4° will cause the eggs to slide.

2. $v = 96.6 \text{ km/h} \times \frac{1000 \text{ m}}{1 \text{ km}} \times \frac{1 \text{ h}}{3600 \text{ s}} = 26.8 \text{ m/s} \checkmark$

$$a = \frac{v^2}{r} = \frac{(26.8 \text{ m/s})^2}{7.6 \text{ m}} = 94.5 \text{ m/s}^2 \checkmark$$

$$a = \left(\frac{a}{g} \right) g = \left(\frac{94.5 \text{ m/s}^2}{9.8 \text{ m/s}^2} \right) g = 9.6g \checkmark$$

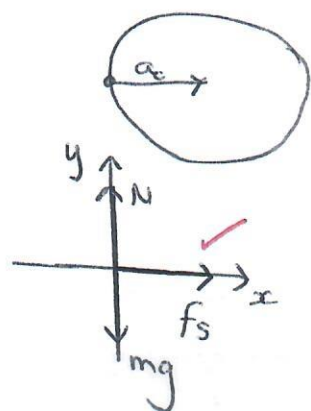
1. A cyclist travels in a circle of radius 25.0 m at a constant speed of 9.00 m/s. The cyclist's mass is 85.0 kg. Draw the free-body diagram and calculate :
(a) the magnitude of the force of friction on the bicycle from the road.
'n Fietsryer ry in 'n sirkel met 'n radius van 25.0 m teen 'n konstante spoed van 9.00 m/s. Die fietsryer het 'n massa van 85.0 kg. Teken die vryligaam-diagram en bereken die grootte van die wrywingskrag op die fiets wat deur die pad uitgeoefen word.
2. A 68 kg crate is dragged across a floor by pulling on a rope attached to the crate and inclined 15° above the horizontal. (a) If the coefficient of static friction is 0.60, what minimum force magnitude is required from the rope to start the crate moving? (b) If $\mu_k = 0.35$, what is the magnitude of the initial acceleration of the crate?
'n 68 kg krat word oor die vloer getrek deur 'n tou daaraan vas te maak en dit teen 15° bo die horisontaal te trek. (a) As die statiese wrywingskoëffisiënt 0.6 is, wat is die grootte van die minimum krag op die tou wat nodig is om die krat te laat beweeg? (b) Indien $\mu_k = 0.35$, wat is die grootte van die aanvanklike versnelling van die krat?

Tutorial Test 5-6

1. $r = 25.0 \text{ m}$

$v = 9.00 \text{ m/s}$

$m = 85.0 \text{ kg}$

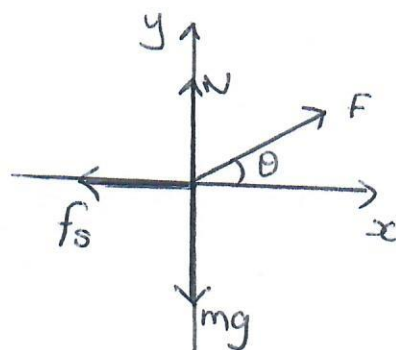


$$\Sigma F_x = ma_c = f_s = m \frac{v^2}{r} \quad \checkmark$$

$$f_s = 85.0 \text{ kg} \left(\frac{(9.00 \text{ m/s})^2}{25.0 \text{ m}} \right)$$

$$= 275 \text{ N} \quad \checkmark$$

2.



$$\Sigma F_y = ma_y = 0 = N + F \sin \theta - mg \quad \checkmark$$

$$N = mg - F \sin \theta \quad \checkmark$$

$$\Sigma F_x = ma_x = 0 = F \cos \theta - \mu_s N \quad \checkmark$$

$$= F \cos \theta - \mu_s mg + \mu_s F \sin \theta$$

$$F = \frac{\mu_s mg}{(\cos \theta + \mu_s \sin \theta)}$$

$$= \frac{0.6 (68 \text{ kg}) (9.8 \text{ m/s}^2)}{\cos 15^\circ + 0.6 (\sin 15^\circ)}$$

$$= 356 \text{ N} \quad \checkmark$$

b) $\Sigma F_x = ma_x = F \cos \theta - \mu_k N = F \cos \theta - \mu_k (mg - F \sin \theta) \quad \checkmark$

$$a_x = \frac{356 \text{ N} \cos \theta - 0.35 [(68 \text{ kg}) (9.8 \text{ m/s}^2) - 356 \sin 15^\circ]}{68 \text{ kg}}$$

$$= 2.1 \text{ m/s}^2 \quad \checkmark$$